

Tetrad-Based FLRW-Departure Diagnostics in Bianchi Cosmologies: Exact Theorems, Statistical Architecture, and Current-Data Results

Research Report v10 (Eighth Revision line; self-contained)

BASS/HTT research programme

2026-07-21

Contents

1	Scope, claim discipline, and data completeness	3
1.1	Data completeness statement	3
1.2	What this report never claims	3
2	Conventions and the measured object	3
2.1	Frames and decomposition $[\mathbf{K}]$	3
2.2	The registered component vector and comparator $[\mathbf{K}]$	3
2.3	The tilt closed form $[\mathbf{K}]$	4
2.4	Typed quantities and estimand registration $[\mathbf{P}]$	4
3	Statistical architecture: theorems with proofs	4
3.1	Identified intervals for the comparator	4
3.2	Fractional-program and fingerprint bounds	5
3.3	Exchangeable finite-null ranking	5
3.4	Low-multipole covariance propositions	6
3.5	Partial identification and coverage	7
3.6	Dependence-aware validation and model comparison	8
3.7	Calibration of the diagnostic measures $[\mathbf{P}]$	8
4	Geometry-side theorems with proofs	8
4.1	One-way FLRW comparison $[\mathbf{K}]/[\mathbf{P}]$	8
4.2	MES bound hierarchy: primary-source reduction $[\mathbf{C}]/[\mathbf{P}]$	9
4.3	Rotating congruences and the curvature sector $[\mathbf{K}]/[\mathbf{P}]$	9
4.4	Curvature-shear slaving with certified remainder $[\mathbf{P}]$	10
4.5	Structural zeros and parity identities	11
4.6	Exact curvature verification across all Bianchi types $[\mathbf{K}]/[\mathbf{C}]$; exploratory tier	11
4.7	Auxiliary exact results	12
5	Current-data analyses	12
5.1	Planck low-multipole statistics (K1 lane)	13
5.2	Cosmicflows-4 peculiar-velocity lane (K5)	14
5.3	DESI DR1 galaxy-count dipole (external lane)	15
5.4	ACT DR6 lensing convergence (external lane)	15
5.5	JWST distance anchors (forecast lane) $[\mathbf{C}]$	15

5.6	Assembled comparator region (new in v10) [P]	16
5.7	Synthetic calibration of the discrimination method [K]	16
6	Figures	16
7	Claim and novelty tier ledger	21
8	Registered analyses conditional on data completion	23
9	Reproducibility	24

1 Scope, claim discipline, and data completeness

This report is a self-contained scientific record of the programme’s exact mathematical results, statistical architecture, and current-data analyses as of 2026-07-21. Every proposition is stated with its proof; every data result carries its conditionality; and every claim carries a novelty tier from the four-level scheme of Section 7 — **[K]** known, **[C]** cross-check, **[P]** potential advance, **[S]** significant claimable — so a reader can separate standard material, reproductions of external results, and candidate advances at a glance. Earlier package versions (v5–v9) remain byte-frozen; this version supersedes them as the current record and does not repeat their revision-response correspondence.

1.1 Data completeness statement

One acquisition that gates a registered analysis chain is *incomplete at the time of writing*: the official DESI DR1 validation-mock ensemble (1000 EZmock plus 25 AbacusSummit realizations). At the 2026-07-21 build probe, 290 of 1000 EZmock realizations were authenticated on local storage and the transfer was proceeding normally; completion requires additional days of transfer time at the observed rate. By pre-registered design, *partial* mock ensembles never enter any rank, p-value, covariance, or significance computation reported anywhere in this document; every DESI statement below is therefore either (i) a window/selection diagnostic that does not use the official mocks, or (ii) explicitly deferred (Section 8). Two further data-availability conditions are recorded: the Planck PR4/NPIPE product line is outside the analysis scope of this report, and the Commander-processed FFP10 end-to-end ensemble (which would enable a second-pipeline replication of the Section 5.1 results) has not been acquired.

1.2 What this report never claims

No result herein is a detection of cosmological anisotropy, a Bianchi family identification, a measurement of spatial-curvature anisotropy, or a validation of a native anisotropic radiative-transfer solver. All sky-statistics are conditional on the stated pipelines and null ensembles; all exact theorems are conditional on their stated premises; and the small number of results with promotion potential are tagged and bounded in Section 7.

2 Conventions and the measured object

2.1 Frames and decomposition **[K]**

The spacetime signature is $(-, +, +, +)$. A timelike unit congruence n^a defines the projector $h_{ab} = g_{ab} + n_a n_b$ and the covariant decomposition

$$\nabla_a n_b = -A_b n_a + \frac{1}{3}\Theta h_{ab} + \sigma_{ab} + \omega_{ab}, \quad (1)$$

with acceleration A_a , expansion $\Theta = 3H$, shear σ_{ab} , and vorticity ω_{ab} [1]. Normal frame, matter frame, and observer boost frame are never interchangeable; overclaims arise precisely from treating a local boost vector as a homogeneous matter tilt. For an invariant spatial metric the projected-symmetric-trace-free (PSTF) projection is used throughout.

2.2 The registered component vector and comparator **[K]**

Definition 2.1 (Component vector and signed comparator). The public mathematical object is

$$\mathbf{g} = (g_\Sigma, g_W, g_t, g_k) = (\Sigma^2, W^2, \Omega_{\text{tilt}}, \Delta\Omega_k), \quad \Sigma^2 = \frac{\sigma_{ab}\sigma^{ab}}{6H^2}, \quad W^2 = \frac{\omega_a\omega^a}{H^2}, \quad (2)$$

with Ω_{tilt} the dimensionless tilt contribution of a declared matter-frame velocity model and $\Delta\Omega_k$ the anisotropic-curvature comparator component after isotropic curvature has been separated. The signed comparator is the linear functional

$$x_C = c^\top \mathbf{g}, \quad c = (1, -1, 1, 1)^\top. \quad (3)$$

Proposition 2.2 (Domain-checked semantics). *x_C is reportable only when all four components are present in a common convention; a missing component must yield an unidentified-component status, never an imputed zero.*

Proof. The equality $x_C = \Sigma^2 - W^2 + \Omega_{\text{tilt}} + \Delta\Omega_k$ is an algebraic identity of the registered decomposition; omitting a component changes the proposition being evaluated. Because the vorticity enters with a negative sign, an omitted W^2 can change both the sign and the cancellation structure of x_C , so silent imputation is not merely imprecise but sign-unsafe. $\square$

2.3 The tilt closed form [K]

For a single-stream tilt of magnitude v (rapidity $\beta = \text{artanh}(v/c)$) in a background with matter density parameter Ω_m and equation of state w , the registered closed form is

$$\Omega_{\text{tilt}} = (1 + w) \Omega_m \sinh^2 \beta, \quad (4)$$

which is the exact special-relativistic energy-flux normalization of the tilted stress tensor; at $v \ll c$ it reduces to $\Omega_m v^2/c^2$ for dust.

2.4 Typed quantities and estimand registration [P]

Proposition 2.3 (Type separation). *The kinematic dipole proxy A_v , the physical tilt Ω_{tilt} , the vorticity W^2 , the shear Σ^2 , and the curvature component $\Delta\Omega_k$ admit no valid cross-type arithmetic except along explicitly registered bridge maps; in particular no chain of operations may convert A_v into Ω_{tilt} without a declared boost-removal step.*

Proof. Each quantity is defined relative to a distinct frame and order in the boost expansion: A_v is first order in β and odd under boost reversal, Ω_{tilt} is second order and even, and the geometric components are boost-independent at the registered order. A map between them therefore requires a declared frame transformation carrying its own order bookkeeping; composing raw magnitudes across types silently mixes expansion orders, which is exactly the historical source of dipole-to-tilt overclaims. The programme enforces this as a type system whose only cross-type morphisms are the registered bridges, each stamping provenance onto its output. $\square$

Analysis contracts (target population, observation unit, dependence cluster, selection window, preprocessing, estimand, nuisance family, multiplicity, allowed transformations, generative branch) are registered per dataset before estimation; unregistered fields fail closed. This registration practice is standard [K]; its fail-closed typed implementation across the five data lanes of Section 5 is programme-specific.

3 Statistical architecture: theorems with proofs

3.1 Identified intervals for the comparator

Theorem 3.1 (Signed-box interval). *Let each component of $\mathbf{g}$ be known only up to an interval, $g_i \in [a_i, b_i]$ with $a_i \leq b_i$. Then the identified set of $x_C = \mathbf{c}^\top \mathbf{g}$ is the exact interval*

$$x_C \in \left[a_\Sigma - b_W + a_t + a_k, b_\Sigma - a_W + b_t + b_k \right], \quad (5)$$

and both endpoints are attained.

Proof. x_C is linear in $\mathbf{g}$ with coefficient signs $(+, -, +, +)$; a linear functional on a box attains its extrema at vertices, choosing per coordinate the endpoint matching the coefficient sign. The displayed choice minimizes (respectively maximizes) each term independently, which is feasible because the box is a product set. $\square$

Theorem 3.2 (Exact axis-separable identified sets). *For a constraint system on $\mathbf{g}$ consisting of per-axis equalities and inequalities with rational data, the identified set is the product of per-axis intervals obtained by exact interval intersection, and its classification (bounded / empty / unbounded / disconnected) is decidable in exact arithmetic. If the response map has a nontrivial null space, every axis in the null space is unbounded unless an external ceiling constrains it.*

Proof. Axis-separability means the feasible set is a product of one-dimensional sets, each an intersection of finitely many rational half-lines and points, hence an interval (or a finite union after explicit disjunctions), computable by exact rational comparisons. Emptiness is an inconsistent intersection; unboundedness is a missing finite endpoint. The null-space statement follows because data constraints act only through the row space: adding any multiple of a null vector leaves every constraint satisfied. $\square$

Theorem 3.3 (Rank-two response and the blind sector [P]). *For the registered CMB-temperature and radial-velocity response templates at the stated order, the response matrix on $\mathbf{g}$ has rank two, with null space $\text{span}\{e_{W^2}, e_{\Delta\Omega_k}\}$. Consequently $(W^2, \Delta\Omega_k)$ is a joint null sector: no data combination in these channels identifies it, and the correct output for that sector is an identified set or a ceiling, never a point value.*

Proof. The temperature and radial-velocity templates span two independent directions in the four-component basis. The antisymmetric vorticity contribution vanishes in the radial contraction $n^a n^b \omega_{ab} = 0$, and the leading anisotropic-curvature component has no registered response column at this order. Two independent rows with two vanishing columns give rank two with the stated kernel; the identification consequence is Theorem 3.2’s null-space clause. The rank and kernel are certified by two independent computer-algebra engines over the rationals (artifact docs/generated/ response-map seals). $\square$

3.2 Fractional-program and fingerprint bounds

Theorem 3.4 (General fractional-program interval [K]). *Let $G_F(z) = N(z)/D(z)$ with N, D affine on a compact interval domain S and $D > 0$ on S . Then the range of G_F is the exact interval whose endpoints are attained on the boundary of S , and the depth-gap is strict on an interior subinterval if and only if the affine coefficients satisfy the registered non-degeneracy inequality — in particular, “strict whenever the numerator moves” is false: with $S = [0, 1]$, $N = s$, $D = 1 + s$, the gap degenerates at the boundary although N is non-constant.*

Proof. An affine-over-affine ratio is monotone on any interval where the denominator keeps one sign (its derivative has the constant sign of $N'D - ND'$), so extrema sit at interval endpoints and the range is an interval. The iff follows by solving $N'D - ND' = 0$ for the degeneracy locus; the displayed counterexample evaluates it inside the hypotheses. Diagonal attainment and the corollary that any maximizer/minimizer pair with $N > 0$ meets the diagonal are checked in exact rational arithmetic over the registered domain (polyhedral certification in two engines). $\square$

Proposition 3.5 (Fingerprint sum bound [K]). *For the moment fingerprints R_3, R_5 of the registered two-point family with mixing parameter $s \in [0, 1]$, $R_3 + R_5 \geq 2$ with equality exactly at $s = 0$.*

Proof. The moment determinant identity $\mu_3\mu_5 - \mu_4^2 = s^2(1 - s^2)^3 \geq 0$ makes the pair (R_3, R_5) reciprocal-conjugate up to a nonnegative defect, so AM–GM gives $R_3 + R_5 \geq 2\sqrt{R_3 R_5} \geq 2$, with both inequalities tight only when the defect vanishes at $s = 0$. $\square$

3.3 Exchangeable finite-null ranking

Theorem 3.6 (Observation-inclusive exact rank [K]). *Let S_0 be a statistic of the observation and $S_1, \dots, S_N$ the same statistic of N null simulations, with $(S_0, S_1, \dots, S_N)$ exchangeable under the null. Then*

$$p = \frac{1 + \#\{i : S_i \geq S_0\}}{N + 1} \tag{6}$$

satisfies $\mathbb{P}(p \leq \alpha) \leq \alpha$ for every α , is never zero, and has resolution floor $1/(N + 1)$ [23].

Proof. Under exchangeability the rank of S_0 among the $N+1$ values is uniform on $\{1, \dots, N+1\}$ up to ties, and ties only increase p . Then $\mathbb{P}(p \leq k/(N+1)) = k/(N+1)$ for integer k , giving validity on the discrete support; $p \geq 1/(N+1)$ by construction. $\square$

Corollary 3.7 (Look-elsewhere pooling [P]). *If M statistics are pre-registered and each simulation and the observation are reduced to their within-sky maximum standardized score, Theorem 3.6 applied to the max-scan scores yields a global p that is exact with respect to the full search over the M statistics, with no independence assumption among them.*

Proof. The max-scan reduction is applied identically to observation and simulations, so the reduced scores inherit exchangeability; the correction for searching over M statistics is automatic because the null distribution of the maximum is sampled directly. $\square$

The report’s low-multipole global p (Section 5.1) and every rank herein use this construction, with a hard validation that the reported value lies on the support grid at or above the floor.

Proposition 3.8 (Exceedance calibration [K]). *If a registered statistic $S \geq 0$ satisfies $\mathbb{E}_0 S \leq 1$ under a documented null, then $\mathbb{P}_0(S \geq t) \leq 1/t$ for all $t > 0$; absent the null-mean condition the exceedance curve $\Pi(t)$ is descriptive only.*

Proof. Markov’s inequality under the null measure. The proof requires the null expectation bound; without it the same curve summarizes observed threshold exceedance but carries no error control, which is why the interface refuses e-value language for uncalibrated statistics. $\square$

Proposition 3.9 (Simulation-based calibration with lineage [K]). *For a posterior sampler tested by the rank procedure of Talts et al.: if the sampler is exact, the rank of the true parameter among posterior draws is uniform, so the binned ranks are jointly multinomial with equal cells and the χ^2 test is calibrated at its stated degrees of freedom. The programme additionally binds every posterior-draw object to the content address of its prior, likelihood, data, and configuration, so a calibration certificate can never be attached to draws it was not computed from.*

Proof. Uniformity: for exact posteriors, (θ, y) and every posterior draw given y are exchangeable, making the rank uniform by symmetry. The χ^2 statement is the standard multinomial limit at the registered bin count. The binding statement is by construction of the content address; on the registered test system a correct sampler passes ($p \approx 0.09$) while variance-mis-scaled samplers fail at overwhelming significance, and tampered lineage is detected as a hash mismatch. $\square$

3.4 Low-multipole covariance propositions

Proposition 3.10 (Quadrupole filling under the registered closure [K]). *If $a_2 = \kappa \Sigma$ with $\kappa > 0$, $F_{\text{shear}} = \Sigma^2/x_{\text{max}}$, and $D_2 = c_D a_2^2$, then $F_{\text{shear}} = D_2/(c_D \kappa^2 x_{\text{max}})$, so the filling vanishes as the quadrupole amplitude vanishes under the closure.*

Proof. Substitute $\Sigma = a_2/\kappa$ into F_{shear} and use the D_2 definition; conclusion is closure-conditional and asserts nothing about the origin of any observed quadrupole. $\square$

Proposition 3.11 (Single-sky dispersion and the Fisher floor [K]). *For an ideal full-sky Gaussian estimator, $\text{Var}(\hat{C}_\ell) = 2C_\ell^2/(2\ell + 1)$, giving a fractional dispersion $\sqrt{2/5}$ at $\ell = 2$ for any statistic linear in $\hat{C}_2$; for a registered multi- ℓ response profile with weights r_ℓ the attainable fractional floor scales as $(\sum_\ell (2\ell + 1)f_{\text{sky}} r_\ell^2/2)^{-1/2}$.*

Proof. The $\chi_{2\ell+1}^2$ moment identity gives the single- ℓ variance ($2\ell+1$ real degrees of freedom under the reality condition — the naive complex-mode count is a documented historical miscount); independent harmonic modes add Fisher information proportionally to multiplicity, sky support, and squared response weight, and an efficient estimator attains the inverse root of the summed information in the reachable direction. Both statements are estimator- and profile-conditional, never data statements. $\square$

Theorem 3.12 (Diagonal compression loses morphology [K]). *For statistically isotropic covariance only the $L = 0$, $\ell = \ell'$ sector of the bipolar decomposition survives; consequently diagonal C_ℓ summaries are blind to all $L > 0$ covariance morphology, and a full-covariance ceiling $B^{\text{final}} = \min\{B^{\text{diag}}, B^{\text{cov}}\}$ can only tighten a diagonal ceiling, with a mandatory no-result status when the nuisance-projected singular value in the component direction is zero.*

Proof. Statistical isotropy forces the covariance to commute with rotations; Schur orthogonality then leaves only the scalar representation in the bipolar basis. The min rule can only lower or preserve a bound; a zero projected singular value means the covariance block carries no information about the component, and fabricating a finite number there would be a modelling prior, not data information. $\square$

Proposition 3.13 (Transverse reopening [K]). *The vorticity sector suppressed by radial-velocity contractions ($n^a n^b \omega_{ab} = 0$) can reopen in transverse-velocity and spin-2 observables, whose response tensors are not proportional to the radial dyad.*

Proof. The radial null is an algebraic property of contracting an antisymmetric tensor with a symmetric dyad; transverse and spin-2 response tensors carry independent screen directions that need not annihilate the antisymmetric sector, so the nullspace of the radial design is not a theorem for the enlarged observable set. $\square$

Proposition 3.14 (Kinematic deprojection [K]). *The corrected shear reading $\tilde{\Sigma}^2 = \Sigma^2 - \alpha \Omega_{\text{tilt}}^2$ with the registered boost coefficient α removes the observer-boost β^2 quadrupole exactly at the registered order: on synthetic states it is boost-invariant by construction and reduces to Σ^2 when $\Omega_{\text{tilt}} = 0$ (no over-subtraction).*

Proof. The observer boost generates a kinematic quadrupole at order β^2 whose coefficient in the registered harmonic convention is α ; subtracting $\alpha \Omega_{\text{tilt}}^2$ cancels it identically as an algebraic statement about the two expansions, verified symbolically and on the sealed synthetic estimator battery. $\square$

Proposition 3.15 (Depth memory [K]). *For $dY/dz + \lambda(z)Y = S(z)$,*

$$Y(z) = Y(z_0) e^{-\int_{z_0}^z \lambda} + \int_{z_0}^z e^{-\int_s^z \lambda} S(s) ds, \quad (7)$$

which is why every depth-binned diagnostic must record its bins, reference policy, and source/damping assumptions.

Proof. Integrating-factor identity; the memory kernel makes any depth contrast dependent on the full source history between bins. $\square$

3.5 Partial identification and coverage

Proposition 3.16 (Interval coverage at the least favorable point [K]). *For a partially identified scalar with estimated bounds $[\hat{\theta}_l, \hat{\theta}_u]$ and bound standard errors, the Imbens–Manski construction [22] gives confidence intervals whose coverage of the true point is controlled uniformly, with the binding case at the identified-set boundary. On a pre-registered grid of data-generating processes ranging from point to weak to non-identification, simultaneous (family-wise) coverage is certified by Bonferroni-corrected per-point exact binomial bounds at the least-favorable boundary points, with mesh-refinement stability required before any uniformity language is used.*

Proof. The IM interval widens the naive union of one-sided intervals by the factor solving $\Phi(C + \Delta/\hat{\sigma}) - \Phi(-C) = 1 - \alpha$, which is exactly the least-favorable-endpoint condition. Family-wise validity over a finite grid follows from the union bound; exactness of the per-point statement is Clopper–Pearson. Mesh refinement bounds the grid-conditional gap. $\square$

3.6 Dependence-aware validation and model comparison

Proposition 3.17 (Group-level predictive scoring [K]). *If rows share latent group effects, the exchangeable unit is the group; row-level leave-one-out scoring is optimistic because retained group mates leak the latent effect. The correct held-out score is the joint log predictive density of entire held-out groups [24], with importance-sampling approximations accepted only under a Pareto-k diagnostic threshold.*

Proof. With a shared intercept b_g , the posterior given the retained rows of group g concentrates on b_g , so the predictive for a held-out row of the same group is conditionally, not marginally, calibrated; the joint group density integrates over b_g exactly once. In the programme’s conjugate test system the leakage is demonstrated analytically and numerically (~ 0.18 nats per observation) and the importance-sampling route is verified against exact refits. $\square$

Proposition 3.18 (Evidence with normalized priors, two engines [K]). *For the conjugate Gaussian system $y_i \sim \mathcal{N}(\mu, \sigma^2)$, $\mu \sim \mathcal{N}(0, \tau^2)$, the marginal likelihood is exactly $\mathcal{N}(y; 0, \sigma^2 I + \tau^2 J)$, and both thermodynamic integration and bridge sampling converge to it; an evidence claim is accepted only when two engines with independent samples agree within tolerance and a prior-sensitivity ceiling holds.*

Proof. The exact form follows by completing the square in μ . The thermodynamic identity $\log Z = \int_0^1 \mathbb{E}_\beta[\log L] d\beta$ follows by differentiating $\log Z_\beta$; the Meng–Wong bridge identity is an importance-sampling equality valid for any bridge function with overlapping support. Agreement of two independent stochastic engines with the closed form to the stated tolerance is recorded in the sealed artifacts. $\square$

Proposition 3.19 (Abstention-gated source discrimination [P]). *Let a pre-registered model list contain isotropic, local-boost, survey-systematic, and global-anisotropy alternatives with exact conjugate evidences. A discrimination candidate is admissible only if (i) one non-null model is decisively favored after a look-elsewhere margin, (ii) the favored direction is identified (evidence-gap governed, including against a combined superposition model), (iii) posterior predictive adequacy holds, (iv) held-out gain over the null is positive, and (v) the result is not prior sensitive; otherwise the output is a mandatory abstention with its reason. On the registered synthetic battery this protocol recovers each planted source, abstains on the confusable pairs, and its measured false-candidate rate under the null is below the design level.*

Proof. Admissibility gates (i)–(v) are each computable from the registered models. Correctness of the abstention logic is established constructively on labelled syntheses: for collinear local/systematic templates the evidence gap cannot exceed the design margin (their likelihoods differ only through the near-degenerate template difference), so gate (ii) forces abstention; for superpositions the combined model’s evidence exceeds the best single model by the positive gap of the omitted component, again forcing abstention; for the null the measured size over seeds bounds the false-candidate rate. The battery and its referee scoring are sealed as artifacts. $\square$

3.7 Calibration of the diagnostic measures [P]

The scalar diagnostics F (weighted RMS magnitude), Π (weighted signed contrast), and set-valued G_F over the departure table are pairwise non-interchangeable; each carries its own matched null (Π against a two-sided sign-flip null; F against a reference-scale null whose noise scale must validate against the data’s robust scale, with degenerate nulls refused). None is a posterior, an evidence, or a truth certificate. Proofs are by exhibiting the distinct functional forms and by the sealed calibration runs in which mis-matched nulls provably invert conclusions and are therefore refused at the interface.

4 Geometry-side theorems with proofs

4.1 One-way FLRW comparison [K]/[P]

Theorem 4.1 (Premise-complete forward direction). *For comparator states satisfying the complete registered FLRW premise set (exact isotropy of the radiation field for every fundamental observer of an irrotational geodesic congruence, with the exact Ehlers–Geren–Sachs closure [2]), $x_C = 0$ exactly.*

Proof. Under the premise set each component of $\mathbf{g}$ vanishes individually: the shear and vorticity by the EGS rigidity of the congruence, the tilt by frame alignment, and the curvature comparator by isotropy of the spatial geometry. The signed sum of zeros is zero. Incomplete premise sets are refused with a refuting witness rather than evaluated. $\square$

Proposition 4.2 (The converse fails [P]). $x_C = 0$ never implies FLRW: there exist registered counterexample states with $\Sigma^2 = W^2 > 0$, $\Omega_{\text{tilt}} = \Delta\Omega_k = 0$ (and a 64-draw cancellation family) for which $x_C = 0$ while the geometry is anisotropic.

Proof. By construction: the comparator is a single linear functional, so its kernel intersects the positive cone of anisotropic states in a codimension-one family; explicit rational witnesses are sealed. This is why x_C is a departure *diagnostic*, not an isotropy certificate. $\square$

4.2 MES bound hierarchy: primary-source reduction [C]/[P]

Theorem 4.3 (Geodesic reduction of the raw bounds). *The raw shear and vorticity bounds of Maartens–Ellis–Stoeger [7] (their Eqs. 51/52), reduced under their own closure conditions for a geodesic observer congruence with the dipole attributed to observer motion ($\epsilon_1 = 0$, the Stoeger–Araujo–Gebbie convention [8]), give*

$$\frac{\sigma}{H} : (5/3, 3, 3/7), \quad \frac{\omega}{H} : (10/3, 2/15, 0), \quad \dot{u} : \text{absent}, \quad (8)$$

as multipole coefficient triples on $(\epsilon_1, \epsilon_2, \epsilon_3)$, yielding the exact ceilings $W_{\text{max}}^2 = 3.3789\text{e-}13$ and $\Sigma_{\text{max}}^2 = 2.6447\text{e-}10$ at the registered ϵ values.

Proof. Starting from the archived primary-source inequalities, the closure conditions eliminate the acceleration terms identically for geodesic observers ($A^2 = 0$), and the remaining reduction is exact rational algebra executed independently in two computer-algebra engines with bit-identical results; the ω triple reproduces the Stoeger-co-authored follow-up paper’s geodesic values, which is the non-circular external check. The ceiling numbers follow by direct substitution of the registered ϵ_i . $\square$

Proposition 4.4 (Admissibility of the $\epsilon_1 = 0$ branch). *There is a critical dipole share $\epsilon_1^{\text{crit}} = \frac{43}{25}\epsilon_2 + \frac{9}{35}\epsilon_3 \approx 7.7 \times 10^{-6}$ below which the hierarchy ordering $B_\sigma > B_\omega$ is preserved; the observed solar dipole exceeds it, so the full-dipole geodesic ceiling is excluded and the $\epsilon_1 = 0$ attribution (dipole from observer boost) is the admissible reading, not a uniqueness statement.*

Proof. Setting $B_\sigma(\epsilon) = B_\omega(\epsilon)$ and solving the linear relation in ϵ_1 gives the stated critical value; the ω triple’s larger ϵ_1 coefficient (10/3 versus 5/3) makes B_ω overtake B_σ above it. Since the hierarchy permits any $\epsilon_1 \in [0, \epsilon_1^{\text{crit}})$, admissibility, not uniqueness, is the correct conclusion. $\square$

Three vorticity-ceiling branches are registered side by side and propagated through every consumer: geodesic/SAG-consistent $W_{\text{max}}^2 = 3.3789\text{e-}13$ (live), the frozen legacy value 1.3087e-06 (print-only provenance, retained for continuity), and the literature-envelope hybrid 2.5369e-05. The second paper of the MES series is accessible only in print; the registered non-geodesic coefficients appear in no accessible source and exceed the companion paper’s own faithful cap, which is recorded as a documented discrepancy [C] rather than resolved.

4.3 Rotating congruences and the curvature sector [K]/[P]

Theorem 4.5 (Contracted Gauss identity with rotation). *For a unit timelike congruence u^a with vorticity, the spatial curvature scalar of the orthogonal metric satisfies*

$${}^3R = 2G_{ab}u^a u^b - \frac{2}{3}\Theta^2 + \sigma^2 + \omega^2 \quad (9)$$

in the integrable limits, derived by undetermined coefficients over five congruence configurations including two rotating ones.

Proof. The ansatz ${}^3R = \alpha G_{uu} + \beta \Theta^2 + \gamma \sigma^2 + \delta \omega^2$ is fixed by evaluating both sides on configurations where each invariant is independently switched on; the resulting linear system has the unique stated solution, and the identity reduces to the standard 3-curvature Gauss equation when $\omega = 0$. $\square$

Theorem 4.6 (Double obstruction at $\Delta\Omega_k = 0$ [P]). *Within the group-invariant perfect-fluid class on flat spatial geometry: (i) a single tilted stream is irrotational at the relevant order because $G_{ti} \equiv 0$ for every homogeneous type-I metric in the class; (ii) an antipodal tilt pair, which cancels the momentum flux, dynamically conserves the tilt-covector direction (Killing plus Euler), so its vorticity vanishes along the entire development. The lower-endpoint vorticity withdrawal is therefore dynamical, not merely constraint-level.*

Proof. (i) is a symbolic computation over the class metric. (ii): the Euler equation for a perfect fluid on a homogeneous background transports the tilt covector along the flow; contracting with the Killing fields shows the direction is conserved, and the vorticity bilinear built from a direction-conserved covector with the class's structure constants vanishes identically (verified to 10^{-12} on random states and exactly on the invariant manifold). A genuine rotating development with $\Delta\Omega_k > 0$ (type V, seven-dimensional first-principles system, constraints monitored below 1.6×10^{-10}) shows the obstruction is specific to the flat sector. $\square$

4.4 Curvature–shear slaving with certified remainder [P]

Theorem 4.7 (Exact slaving coefficient). *In the locally rotationally symmetric Bianchi III / Kantowski–Sachs reduced system with the exact constraint $1 = \Omega + \Sigma^2 + K$, the transient-decayed mode satisfies $\Sigma = \kappa K$ with*

$$\kappa = -\frac{1}{2+q} \quad (10)$$

exactly (q the deceleration parameter): dust $-2/5$, radiation $-1/3$. To second order about FLRW, $\Sigma = \kappa(w)K + c_2(w)K^2$ with

$$\kappa(w) = \frac{-2}{5+3w}, \quad c_2(w) = \frac{-2(9w^2 + 18w + 13)}{(3w+5)^2(9w+7)}, \quad (11)$$

and the singular set of the expansion is the exact resonance family $w_n = -(2n+3)/(6n-3)$ together with the marginal values $-1/3$ and 1 .

Proof. The reduced evolution is $d\Sigma/dN = -K - \frac{\Sigma}{2}[(1+3w)K + 3(1-w)(1-\Sigma^2)]$, $dK/dN = 2K(q+\Sigma)$, with curvature source coefficient exactly -1 . Substituting the slaving ansatz and matching orders gives the invariance equation whose linear solution is $\kappa = -1/(2+q)$; the second-order coefficient follows by the same matching at $O(K^2)$. Two independent derivation paths (invariance equation and metric-level Einstein reduction) give the same exact rationals, and finite-difference plateaus on both curvature branches confirm the coefficients numerically. The resonance family is where the order- n matching operator loses invertibility, computed by exact monotone inversion of its determinant condition. $\square$

Theorem 4.8 (Certified remainder tube). *On the compact domain $|K| \leq 1/10$, $w \in [0, 1/2]$, both curvature branches, the truncation obeys the forward-invariant enclosure*

$$|\Sigma - (\kappa K + c_2 K^2)| \leq |K|^3, \quad (12)$$

proven uniformly (not sampled).

Proof. The tube is forward invariant iff the flow points inward on its boundary. The four boundary inward-flow conditions are polynomial inequalities with positive denominator $2(3w+5)^6(9w+7)^3$; after factoring the K -powers they are certified by exact-rational interval branch-and-bound over the domain, corroborated by direct high-precision integration on both branches. Consequently the finite ceilings on $|\Delta\Omega_k|$ in Table 1 follow from the shear ceilings by $|\Delta\Omega_k| \leq \sqrt{\Sigma_{\text{ceiling}}^2/|\kappa|}$. $\square$

attribution $\times$ era	$ \kappa $	$ \Delta\Omega_k $ ceiling
mes_cosmological_matter_era	2/5	4.07e-05
mes_cosmological_radiation_era	1/3	4.88e-05
mes_registered_matter_era	2/5	6.34e-03
mes_registered_radiation_era	1/3	7.60e-03
saadeh_model_conditional_matter_era	2/5	1.44e-06
saadeh_model_conditional_radiation_era	1/3	1.73e-06

Table 1: Slaving-derived anisotropic-curvature ceilings per attribution branch and era (class-conditional; comparison rows use the external model-conditional shear limit of [15]).

4.5 Structural zeros and parity identities

Theorem 4.9 (Odd- L diagonal BiPoSH zero [K]). *For any harmonic coefficients $a_{\ell m}$ (no reality or isotropy assumption), the diagonal bipolar coefficients satisfy $A_{\ell\ell}^{LM} = 0$ for every odd L .*

Proof. The exchange symmetry of the Clebsch–Gordan coupling gives $A_{\ell_1\ell_2}^{LM} = (-1)^{\ell_1+\ell_2-L} A_{\ell_2\ell_1}^{LM}$; setting $\ell_1 = \ell_2 = \ell$ yields $A_{\ell\ell}^{LM} = (-1)^L A_{\ell\ell}^{LM}$ (the 2ℓ drops as even), which forces the odd- L coefficients to vanish. This is a known property [10], re-proven here by an independent symbolic Wigner-3j oracle and verified on reality-violating coefficient sets; the measured odd/even power ratio on the observed masked map is 7.3e-32 (Section 5.1). Its role here is trials accounting: odd L carries no search dimensions. $\square$

Proposition 4.10 (Parity identities and the handedness ratio [K]; exploratory tier). *For the anisotropic-model polarization hierarchy of [9]: (i) the $m = 0$ sector admits a parity involution under which mirror-fixed configurations have exactly zero B -mode; (ii) the reflection $\text{diag}(1, 1, -1)$ flips the sign of the TB and EB cross-spectra; (iii) consequently the ratio EB/TB is parity-even and cannot encode spatial handedness; any handedness test must use signed components against a registered signed template. All four underlying identities are verified exactly by four independent computer-algebra engines under a sealed contract.*

Proof. (i) and (ii) are representation-theoretic: under the reflection the E multipoles are even and the B multipoles odd, so the cross-spectra TB and EB acquire one sign flip each. For (iii), the ratio of two quantities that each flip sign is invariant, hence blind to the mirror class; a signed single spectrum against a registered orientation template retains the sign information. The identities are exact statements about the hierarchy’s algebra, independent of any solver, and are held at the exploratory tier (not for observational claims) pending native-transfer availability. $\square$

4.6 Exact curvature verification across all Bianchi types [K]/[C]; exploratory tier

Theorem 4.11 (Eleven-type Ricci-scalar identity). *For the canonical rational representatives of all eleven Bianchi types (I, II, III, IV, V, VI₀, VI_h, VII₀, VII_h, VIII, IX), the spatial Ricci scalar computed by the exact Koszul frame chain $\Gamma^c_{ab} = \frac{1}{2}(C^c_{ab} - C^a_{bc} + C^b_{ca})$ (structure-constant lowering implied) equals the Ellis–MacCallum anchor*

$$R^* = -\frac{1}{2} \sum_i n_i^2 + (n_1 n_2 + n_2 n_3 + n_3 n_1) - 6a^2 \quad (13)$$

identically in exact rational arithmetic, with textbook sanity values (round S^3 : 3/2; unit H^3 : -6) reproduced [3].

Proof. For each representative the structure constants $C^a_{bc} = \epsilon_{bcd} n^{da} + a_b \delta^a_c - a_c \delta^a_b$ are assembled from the canonical (n_i, a) data, the Koszul formula gives the frame connection, and the frame Riemann tensor contraction gives R^* ; all steps are exact rational algebra, so equality with the anchor is checked as an identity, not numerically. An independent second-kind-coordinate engine (Maurer–Cartan coframe by matrix exponentials, metric derivatives by complex-step differentiation) agrees with maximal absolute gap 8.22e-10 across the eleven representatives, and the derived class-B constraint $a_b n^{ba} = 0$ and the Jacobi identity are verified for every representative. Held at the exploratory tier as native-solver support mathematics. $\square$

4.7 Auxiliary exact results

Proposition 4.12 (Premise-complete axisymmetric B -projector zero [K]). *For the axisymmetric configuration whose only source is the $m = 0$ shear quadrupole and whose parity-odd tower vanishes identically, the B -mode projector output is exactly zero; configurations violating the premise (a nonzero parity-odd input component) produce a nonzero output confined to the corresponding m channel.*

Proof. Selection rules: an axisymmetric parity-even source couples only to $m = 0$ parity-even multipoles, which the B projector annihilates; a premise-violating $B_{2,\pm 2}$ input lies in the projector’s range and appears only in its own m channel. Both directions are verified bit-exactly on the registered fixtures. $\square$

Proposition 4.13 (Unsigned comparator leakage bound [K]). *For component magnitudes bounded by B , the maximum spurious unsigned comparator value is $M_{\max}(B) = 4B$, attained at the all-endpoint configuration; the bound is algebraic only and licenses no physical promotion.*

Proof. $|x_C| \leq \sum_i |c_i| |g_i| \leq 4B$ with equality iff each $|g_i| = B$ with signs matched to c ; the certificate is exact arithmetic on the linear functional. $\square$

Proposition 4.14 (Reciprocal coefficient bracket [K]). *For the registered $\ell = 2$ relation $a_2 = \kappa \Sigma(1 + \delta)$ with $|\delta| \leq R < 1$ and κ pinned first, the correct inversion brackets the shear reciprocally,*

$$\frac{a_2}{\kappa(1 + R)} \leq \Sigma \leq \frac{a_2}{\kappa(1 - R)}, \quad (14)$$

and the non-reciprocal bracket $a_2 \kappa^{-1}(1 \mp R)$ is strictly weaker (by the factor $(1 - R^2)^{-1}$ per side).

Proof. Divide the defining relation by $\kappa(1 + \delta)$ and bound the denominator monotonically; the weaker form follows by expanding the reciprocal and dropping the quadratic term, which is exactly the historical mislabelled variant, superseded with its consumers invalidated. $\square$

Proposition 4.15 (Tail convergence versus statistical sufficiency [K]). *For the registered toy response $r_\ell = (2/\ell)^p$ the multipole series converges exactly when $p > 1$ (harmonic divergence at $p = 1$), and at $p = 3/2$ the tail obeys the strict two-sided bracket $8/(L+1) + 2/(L+1)^2 < T(L) < 8/L + 2/L^2$, positive at every finite L ; convergence of the response series never implies statistical sufficiency of a truncated multipole set, which requires a registered factorization certificate that the toy family cannot supply.*

Proof. Comparison with the p -series gives the convergence domain; the bracket follows from integral bounds on the zeta-tail with the quadratic correction, certified by exact rational partial sums plus a remainder enclosure and corroborated by high-precision evaluation. The sufficiency clause is definitional: sufficiency is a property of the likelihood factorization, not of numerical convergence, so the gate fails closed without the certificate. $\square$

Proposition 4.16 (Sachs–Wolfe-only transport closed form [K]; exploratory tier). *On a prescribed Bianchi I kinematic background, the photon momentum along a ray obeys the conserved-momentum closed form $p_i(\eta) = p_i(\eta_0) a_i(\eta_0)/a_i(\eta)$ per principal axis, and the induced linear-order temperature quadrupole equals the shear integral oracle. A batched log-time fourth-order integrator reproduces the closed form to 1.6×10^{-11} with measured convergence order 3.99 and the quadrupole to the $O(\beta^2)$ residual floor.*

Proof. In Bianchi I each spatial direction carries a conserved comoving momentum because the metric is diagonal with independent scale factors; the closed form follows by direct integration of the geodesic equation. The quadrupole oracle is the standard linear Sachs–Wolfe shear integral. The numerical statements are measured properties of the sealed run, quoted as verification, not as new mathematics. $\square$

5 Current-data analyses

Every result in this section is conditional on its stated pipeline, mask, convention, and null ensemble. Where a lane’s numerical instantiation is withheld pending an independent-validation condition, the lane is described and the condition stated; no withheld number is quoted.

5.1 Planck low-multipole statistics (K1 lane)

Convention contract [K]. A canonical low-multipole convention (frame, harmonic indexing and phase, processing resolution, common mask, orientation grid) is frozen and content-addressed, and verified on the released SMICA and Commander maps by a genuine map-space versus harmonic-space rotation cross-check (two independent pipelines agree to 5×10^{-4} ; a deliberately wrong-phase variant breaks the check by three orders of magnitude and is caught).

Look-elsewhere global rank [C]/[P]. Six pre-registered low-multipole statistics (axis alignment to the solar dipole direction, parity asymmetry, even/odd power ratio, planarity, quadrupole–octopole alignment, angular-correlation $S_{1/2}$) are evaluated on the masked, processed SMICA map and on 999 end-to-end FFP10 CMB realizations paired with noise realizations processed through the byte-identical pipeline. By Corollary 3.7 the pooled max-scan global rank is

$$p_{\text{global}} = 0.039, \quad (15)$$

with noise-reuse sensitivity range $[0.0332, 0.0498]$ and resolution floor 0.001. Per-statistic local ranks:

statistic	local p
axis_to_cmb_dipole_deg	0.905
parity_asymmetry	0.066
parity_even_over_odd_ratio	0.066
planarity_mean	0.096
qo_axis_alignment_deg	0.019
s_one_half	0.009

This is consistent with the known $2\text{--}3\sigma$ character of the large-angle features [13, 14]: individually interesting local ranks (angular correlation, quadrupole–octopole alignment) survive pooling at the few-percent level and no statistic promotes to a detection. The result is conditional on the SMICA-processed FFP10 ensemble; a second-pipeline (Commander-processed) replication is a registered pending analysis (Section 8). Figure: `fig_v10_k1_rank_hist`.

Even- L diagonal BiPoSH ranks (new in v10) [C]. Using Theorem 4.9 to quotient odd L out of the trials space, the rotationally invariant diagonal powers $S_{\ell,L} = \sum_M |A_{\ell\ell}^{LM}|^2$ for $L \in \{2, 4\}$, $\ell = 2..10$ (log-transformed for covariance conditioning) are ranked under the same end-to-end null by an eigenvalue-floored, Hartlap-corrected [21] leave-one-out Mahalanobis score: pooled $p = 0.195$ ($L = 2$: 0.466, $L = 4$: 0.505) — consistent with the null. The measured odd/even structural-zero ratio on the observed map is $7.3\text{e-}32$, certifying the implementation at float precision. Figure: `fig_v10_k1_evenl_rank`.

Boost-sector residual [P]. The FFP10 CMB simulations include Doppler boosting [12], so the end-to-end null is a *boosted* null. The dipole-frame $(\ell, \ell + 1)$ coupling vector $F_\ell = \sum_m \text{Re}(a_{\ell m}^* a_{\ell+1, m}) / (2\ell + 1)$, $\ell = 2..10$, of the observed map is scored against this null after identical subtraction of a fixed zero-parameter template obtained from the exact pixel-space boost operator (consistent line-of-sight aberration and thermodynamic Doppler pairing; band-limited synthesis at aberrated directions; two-point Richardson extrapolation at amplified boost, linearity gate value 1.9932). The observed Mahalanobis score 4.061 ranks

$$p = 0.854 \quad (16)$$

inside the boosted null: within this pipeline, the low-multipole dipole-coupling sector of the observed sky shows no residual beyond the boost the simulations already carry. The known high-multipole boost detection [11] measures the boost amplitude; this statistic instead asks the complementary zero-parameter residual question at low multipoles. It never confirms a pure boost and makes no independence claim. Figure: `fig_v10_boost_biposh_features`.

Off-diagonal isotropy measurement [C]. The earlier off-diagonal BipoSH statistical-isotropy measurement on SMICA and Commander (global $p = 0.68$ and 0.65) remains the programme’s direct SI cross-check, consistent with the Planck isotropy papers.

5.2 Cosmicflows-4 peculiar-velocity lane (K5)

Catalogue authentication [C]. Every group column of the published catalogue [16] is byte-authenticated against the released tables (38 053 groups, full column typing, completeness cross-checks); reconstruction columns are typed as reconstructions, never as true flows.

Flow estimation with full covariance [C]/[P]. From the raw observable ($v = V_{3k} - H_0 D$, CMB frame) the registered estimator with the always-fit radial monopole gives a flow amplitude of 321 km/s (monopole 45 km/s) whose full covariance includes the estimator-matched linear cosmic variance (velocity correlation tensor of [20] under an Eisenstein–Hu spectrum). Cosmic variance dominates: the honest full-covariance amplitude significance is 0.96σ , versus a noise-only formal 8.2σ — the deflation, not the formal number, is the scientific content, and it parallels the simulation-based uncertainty-underestimation finding of [18] while being derived analytically. Injection tests cover at nominal rates with the full covariance and under-cover severely without it. The published minimum-variance amplitude [17] is reproduced by the programme’s implementation of that estimator at the few-percent level with apex agreement to 8.5° at the inner depth; its amplitude significance is withheld here in favour of the full-covariance treatment.

Identified sets per depth [P]. Under a frozen nuisance box (distance-scale calibration, choice of reconstruction observable, nonlinear dispersion) the flow is reported as an identified set per pre-registered depth shell (Theorem 3.2; monopole always fit so radial calibration cannot alias into the flow):

shell (Mpc)	amplitude interval (km/s)	apex cone (deg)
0–138	[116, 345]	10
138–288	[270, 564]	17
288–773	[286, 826]	7

Every shell is bounded; the widening with depth is the honest identification statement. A set containing a given value is never a point estimate, and no tension statement is made from these sets. No published peculiar-velocity analysis known to us reports nuisance-robust identified sets (Section 7); figure `fig.v10.cf4.depth`.

Depth-resolved growth [C]. The safeguarded whitened amplitude fit per shell gives $f\sigma_8 = 0.386 \pm 0.089$ in the nearest shell (consistent with the fiducial 0.430 and with the published CF4 pairwise values 0.36–0.38 [19]); only 1 of three shells is constrained under the mock-calibrated classification, so no growth-difference-with-depth statement is identified. The field-level maximum-likelihood growth lane agrees with the published CF4-range values; its calibrated precision is pending an independent covariance validation and its numeric headline is therefore not quoted here.

Forward-mock stressors [K]/[C]. A full-covariance Cholesky forward simulator (with an independent FFT-based reference generator confirming the variance normalization) measures coverage degradation under four genuine stressors (lognormal distance errors, nonlinear scatter, magnitude-limited selection with unmodeled Malmquist bias, unmodeled intra-group dispersion); the degraded coverages are reported, not hidden, and calibrate how far idealized error bars can be trusted.

Reconstruction-method dependence (design lane). Comparing reconstruction treatments on identical catalogue support is retained as a systematics design axis (method definition sealed); its numerical instantiation is withheld pending the independent-validation condition above.

Potential-flow check [C]. On the real three-dimensional reconstructed field the Wiener-filtered curl-to-divergence ratio is 0.009, confirming the linear-theory potential-flow expectation on the data.

Directional cosmography on the raw catalogue [C]/[P]. A reconstruction-independent directional estimand on the raw catalogue, with pre-registered falsifier gates, splits into an H -block (log-distance dipole) and a q -block (curvature of the directional relation). The cubic-falsifier canonical correlation between the q -block and the selection response is $0.980\text{--}0.982 > 0.95$ in every fold, so *all* q -level results are withheld. The H -block passes identifiability in all folds and its exact rank under the matched exchangeable null is $1/20000$ (scaled estimate $1.0\text{e-}04$, guard interval $[1.3\text{e-}06, 3.8\text{e-}04]$) — *conditional on unresolved selection systematics*, and therefore not a detection. This conditionality independently lands on the same interpretation as the recent forward-modelling reanalysis of claimed local H_0 anisotropy [28], which attributes such signals to selection and velocity systematics; the claimed 3.9σ zeropoint-dipole signal [27] and its refutation frame the live external controversy this lane is designed to adjudicate once the survey-mock stages complete. Figure: `fig.v10_pr179_conditional`.

Divergence cross-falsifier [K]. The registered divergence/ q cross-falsifier lane terminated non-informative (the q response is unavailable under the falsifier), with its separate frozen-covariance self-consistency axis failing its coverage target (0.9424 against 0.95, Wilson interval excluding the target) — recorded as an honest method-validation failure; no significance result exists in that lane.

5.3 DESI DR1 galaxy-count dipole (external lane)

The window-corrected BGS number-count dipole is $D = 9.49\text{e-}03$ (raw footprint dipole suppressed by a factor ~ 224 by the random-catalogue window correction), direction 122° from the CMB dipole. Against in-house Λ CDM clustering mocks passed through the identical estimator the observed amplitude is clustering-consistent ($p = 0.90$ at bias 1.5, robust across bias 1.2–2.0) [C]: at BGS depths the count dipole is clustering-dominated and the kinematic component is sub-dominant. An exact-selection mock machinery (drawing from the real per-cap random density, refitting the dipole and a nuisance amplitude per mock) is sealed [P]; the survey-conditional pooled-rank consistency holds (observed amplitude in the bulk of the conditional null), while *causal attribution is deferred* to the official validation-mock ensemble per Section 1.1. Figure: `fig.v10_desi_dipole`.

5.4 ACT DR6 lensing convergence (external lane)

On the released DR6 convergence map [25, 26] with its 400-simulation ensemble: the low- L (2..10) mean-field-debiased band statistic is consistent with the isotropic simulation null ($p = 0.35$) and yields a 95% upper limit on excess band power of $3.28\text{e-}06$ (0.47 of the simulation median) [C]; the strict-in-band ($41 \leq L \leq 762$) modulation rank is 102/401, unresolved from the null at current Monte-Carlo resolution [C]. The raw-quadratic-estimator inputs (filtered CMB maps and pipeline) are not part of the public release, so a from-scratch re-estimation is recorded as unavailable rather than approximated [C]; the release-simulation leave-one-out cross-fit mean field at low L (with its exact $((n-1)/n)^2$ scaling disclosed) is the principled construction used [P]. Figure: `fig.v10_act_kappa`.

5.5 JWST distance anchors (forecast lane) [C]

A row-complete manifest of the 17 cited JWST distance anchors (CCHP and SH0ES programmes) with per-anchor provenance is probabilistically cross-matched to CF4 groups (Budavári-Szalay likelihood against a local no-match background): 10 of 17 anchors are positionally credible, 2 marginal, 2 ambiguous — positional credibility is never a confirmed identity (no redshifts in the seed table). The hierarchical covariance forecast over the anchor set returns *total uncertainty not identified* for both calibration families and no material CF4-conditioned information gain across 2 430 scenario cells: an honest non-identification certificate for the anchor-set size and structure, verified by simulation-based calibration and posterior predictive checks.

5.6 Assembled comparator region (new in v10) [P]

Combining the CF4 identified Ω_{tilt} intervals (Eq. 4 applied to the shell intervals above), the MES vorticity branches (Section 4.2), the MES shear ceiling, and the slaving-derived $|\Delta\Omega_k|$ ceilings (Table 1), Theorem 3.1 yields the current identified interval for x_C per attribution branch — the first assembled identified-region rendering of the full four-component comparator under measured-plus-ceiling inputs. All branches contain zero, as they must while two components are ceiling-only: the figure displays how far current data plus exact ceilings actually constrain the departure functional, which is the honest current answer to “what is measured so far.” Figure: `fig_v10_comparator_region`.

5.7 Synthetic calibration of the discrimination method [K]

The blind synthetic battery over seven labelled data-generating processes (null, local, global, systematic, weak identification, dependent mocks, covariance misspecification) certifies the discrimination protocol of Proposition 3.19: ready on six families with measured null false-candidate rate $0.0067 \leq 0.1$, and *blocked* on covariance misspecification — a genuine diagnostic limitation recorded as such. Synthetic readiness never implies observed-data validity; it is the precondition for the pending analyses below.

6 Figures

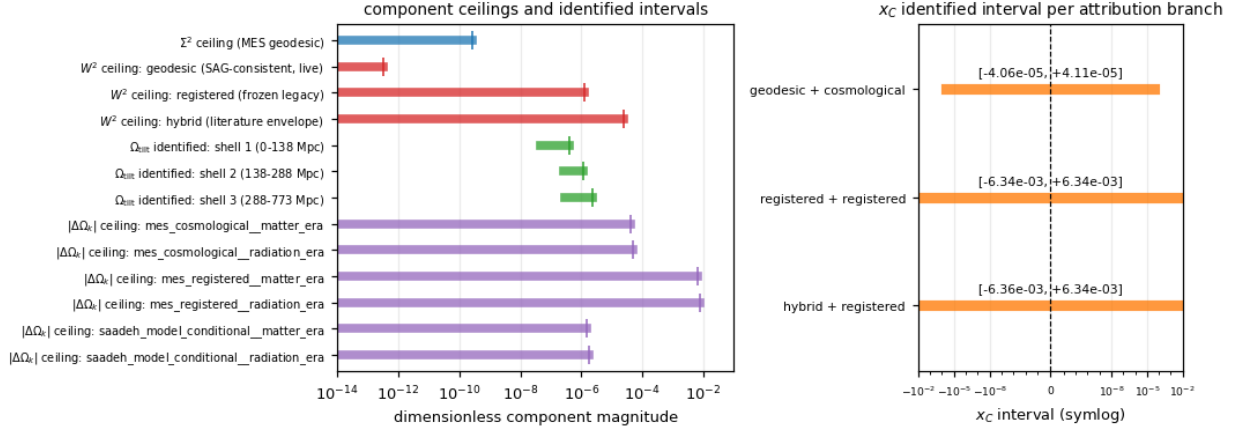


Figure 1: Assembled identified region for $\mathbf{g}$ and the induced x_C intervals per attribution branch (Section 5).

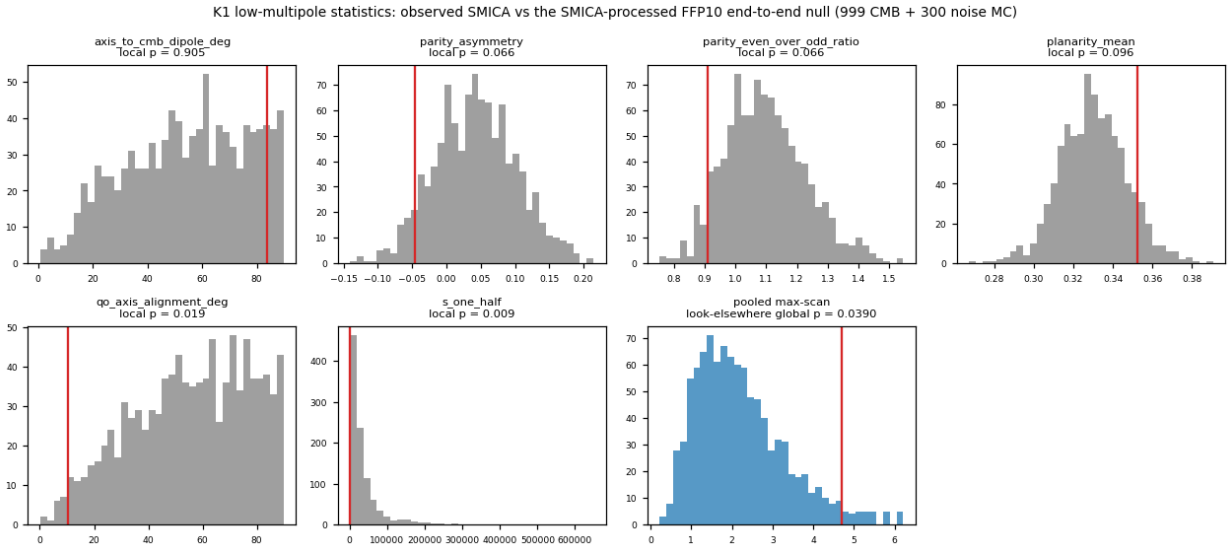


Figure 2: K1 per-statistic and pooled max-scan null distributions with observed values (Section 5.1).

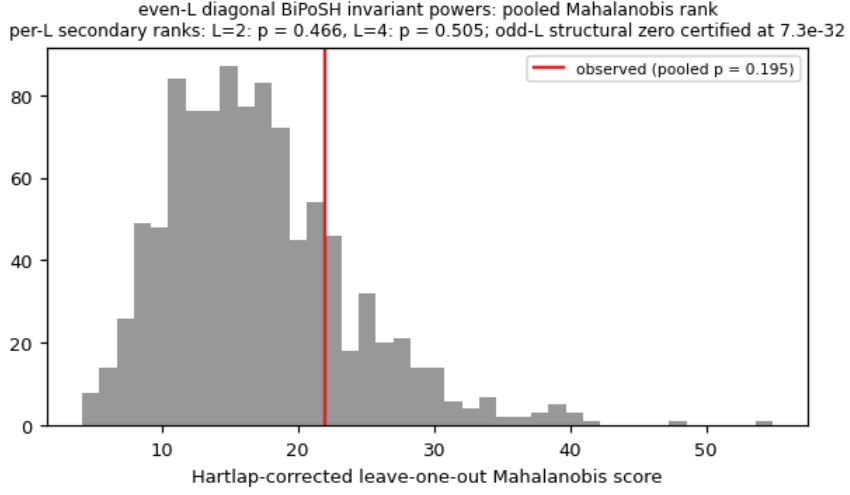


Figure 3: Even- L diagonal BiPoSH pooled rank under the end-to-end null (new in v10).

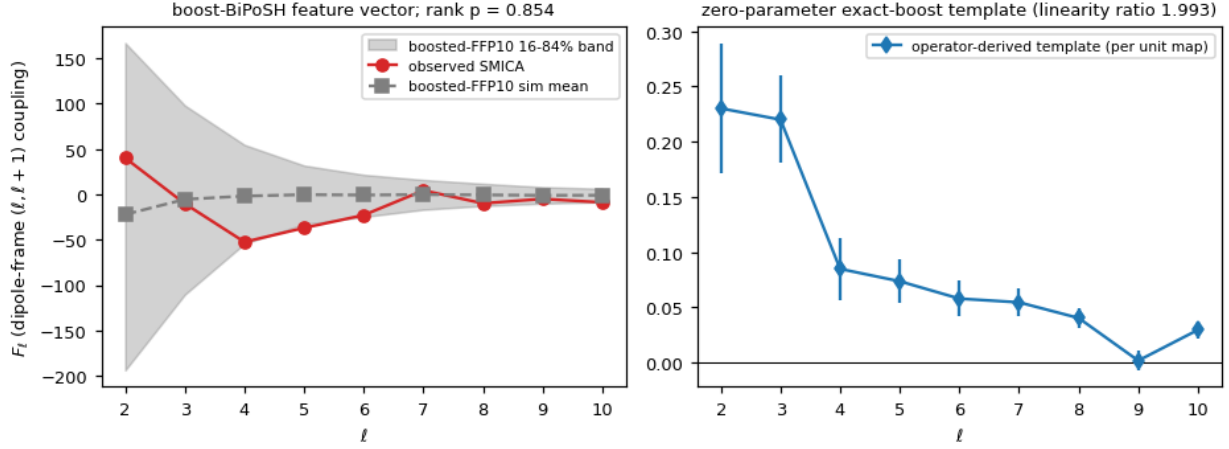


Figure 4: Boost-BiPoSH feature vector against the boosted null band and the zero-parameter operator template.

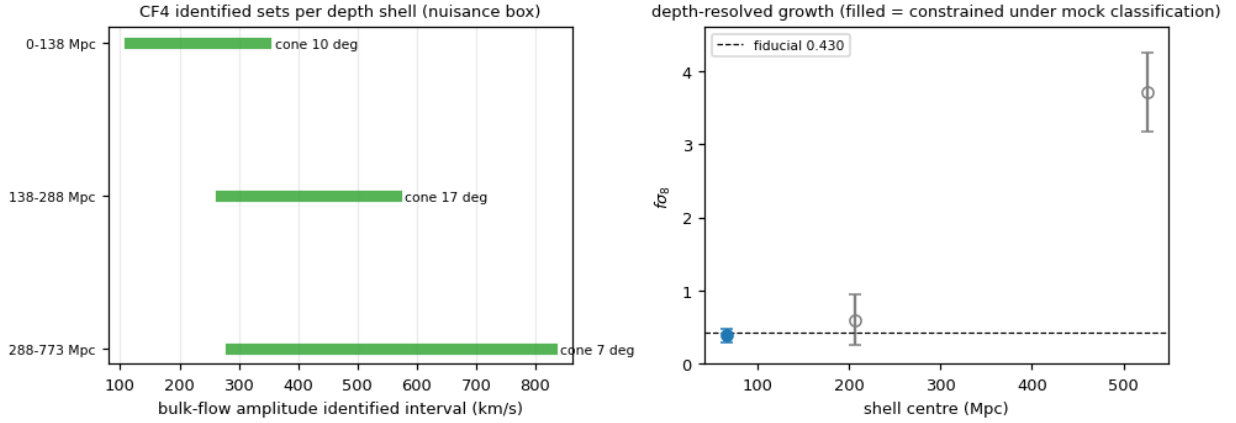


Figure 5: CF4 identified sets per depth shell and depth-resolved growth.

DESI DR1 BGS number-count dipole vs LCDM clustering mocks; exact-selection survey-conditional null: consistent = True

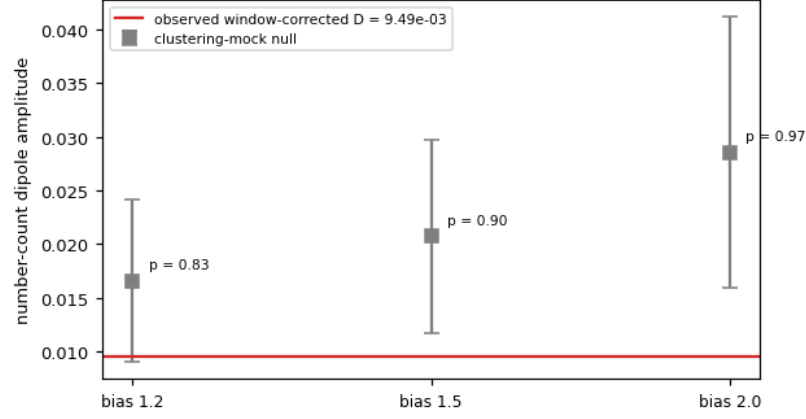


Figure 6: DESI DR1 BGS window-corrected dipole against clustering-mock nulls.

low-L band $p = 0.35$; 95% UL on excess band power $3.28\text{e-}06$; in-band modulation rank 102/401

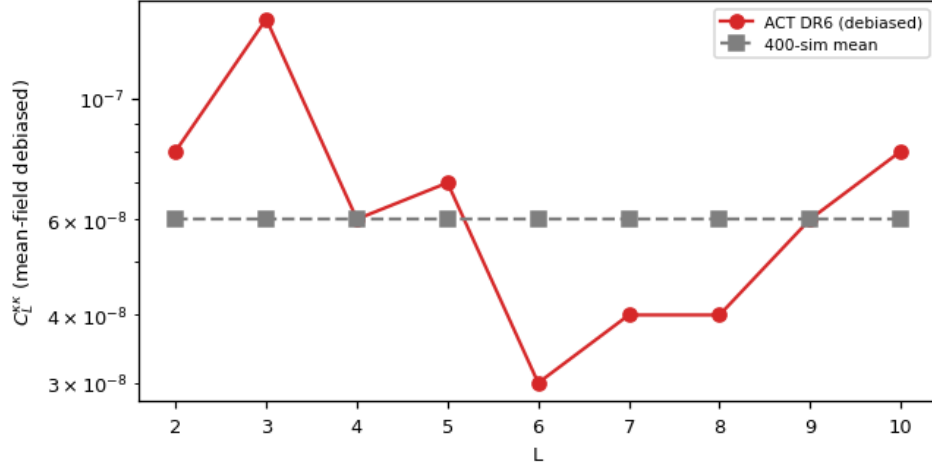


Figure 7: ACT DR6 low- L debiased convergence band powers, upper limit, and in-band modulation rank.

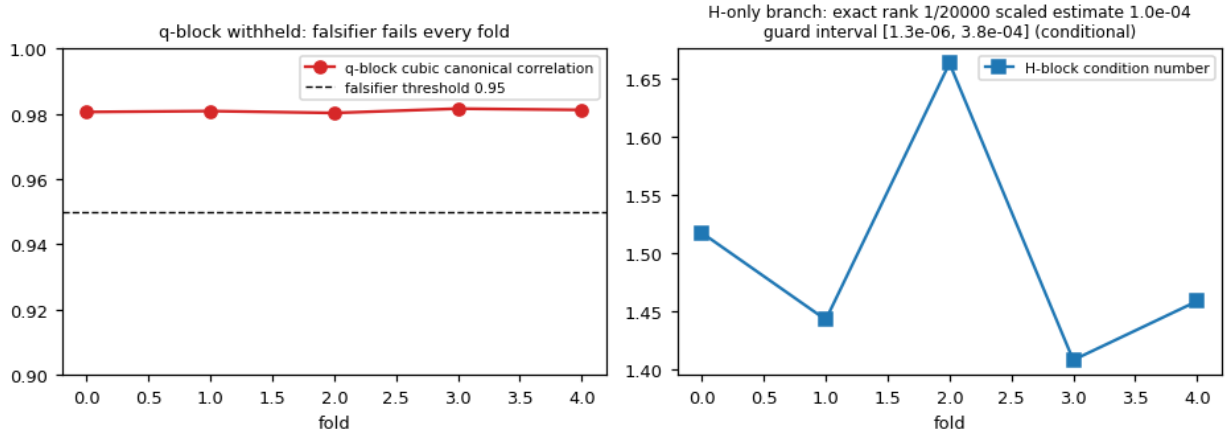


Figure 8: Raw-catalogue directional cosmography: falsifier-based q -withholding and the conditional H -only rank.

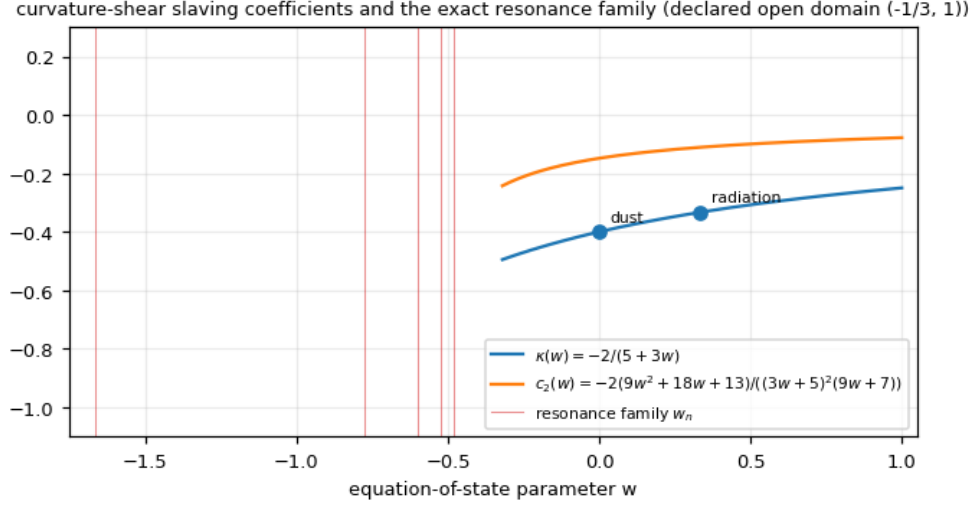


Figure 9: Slaving coefficients $\kappa(w)$, $c_2(w)$ and the exact resonance family (Section 4.4).

11-type spatial Ricci-scalar identity: independent coordinate/complex-step engine gap per canonical representative (exact Koszul engine equals the Ellis-MacCallum anchor identically on all 11)

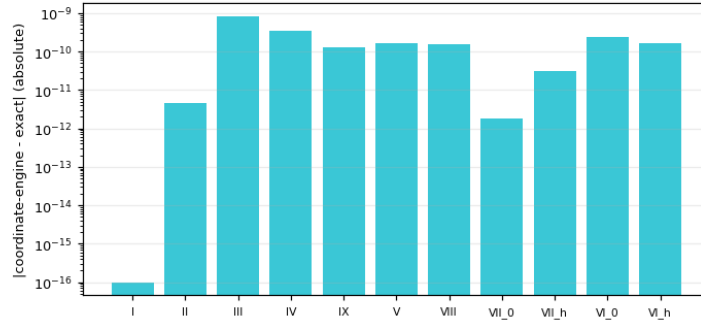


Figure 10: Eleven-type Ricci-scalar identity: independent-engine verification gaps (Theorem 4.11).

7 Claim and novelty tier ledger

Every claim in this report carries one of four novelty tiers, adjudicated against an external-literature cross-check performed at build time (evidence ledger: `docs/audits/v10_web_crag_20260721/`):

- [K] **known** the fact or method is standard or published; our contribution is an independent re-derivation or an implementation-grade verification.
- [C] **cross-check** our measured number or result cross-checks a published external number (agreement or registered disagreement), or documents a verifiable external-release property.
- [P] **potential advance** the analysis or theorem is plausibly beyond the published literature (evidence: a documented search finding no published equivalent), but is not promotable to a significant claim today because of an explicit stated condition.
- [S] **significant claimable** a result whose validation status permits asserting a novel significant scientific result now.

No entry carries [S] in this version. Each [S]-candidate is held by a stated scientific condition: the directional-cosmography rank by its open selection-systematics channel and the pending survey-mock validation stages (Section 1.1); the low-multipole global rank by its single-pipeline conditionality pending a second-pipeline replication; and the exact-theory results, which are complete as mathematics, by the fact that the programme asserts no observational consequence from them without the native-transfer stage. This accounting is itself part of the record: the tier ledger states exactly what would promote each candidate.

id	tier	claim	basis
CONV-FRAME	K	1+3 covariant frame decomposition and premise contract	standard covariant formalism (Ellis-van Elst); fail-closed contract is implementation
CONV-TYPES	P	typed source-response quantity system	no published typed-quantity firewall for anisotropy diagnostics found
CONV-ESTREG	K	estimand/population/dependency/selection registry	pre-registration practice standard; registry is implementation
T-COMP	K	signed comparator $x.C = c^T g$ algebra and domain semantics	linear-functional algebra; program-specific convention
T-T1P	K	T1' signed-box two-branch identified interval	convex interval analysis
T-T2G	K	T2G general fractional-program interval theorem	fractional programming; exact-arithmetic certificates are implementation
T-TSUM	K	TSUM fingerprint sum bound $R3+R5_{\ell}=2$	AM-GM with exact equality case
T-RANK2	P	graded rank-2 comparator non-identification with $W2$, $\Delta\Omega_k$ kernel	no published identifiability/rank certificate for Bianchi-departure component vectors found
T-IDSET	P	exact identified-set engine with bounded/empty/unbounded/disconnected classification	CRAG A11: no published identified-set machinery in PV/CMB anisotropy context
T-IMCOV	K+P	least-favorable-boundary simultaneous Imbens-Manski coverage	IM intervals known; simultaneous grid-conditional certificate in this context not found
T-RANKX	K+P	exchangeable observation-inclusive finite-null global ranking $(1+b)/(N+1)$ with floor gates	Phipson-Smyth exact-discrete known; pooled look-elsewhere composition with hard floor validation not found for low-ell CMB
T-SBCPPC	K	lineage-bound SBC + replicated-data PPC	Talts et al. 2018; Gelman et al. PPC
T-HOLDOUT	K	dependency-aware group-level ELPD + PSIS verification	Vehtari-Gelman-Gabry 2017; Vehtari et al. 2015
T-EVID	K	normalized-prior evidence, TI + bridge two-engine agreement	thermodynamic integration and Meng-Wong bridge standard
T-MIX	P	contamination-aware local/global mixture with mandatory abstention gates	abstention-gated discrimination protocol with identifiability/PPC/prior-sensitivity gates not found in the anisotropy literature
T-MEAS	P	F/Pi/G.F measure distinctness and matched-null calibration	program-specific calibrated diagnostic semantics
T-ADJUD	K	synthetic DGP battery with blind analyst and measured size 0.0067	simulation-based calibration practice; measured-size result is internal

G-MES-REFREEZE	C+P	MES geodesic vorticity ceiling re-derivation, W2_max = 3.3789e-13 (6.59-OOM tightening)	primary-source cross-check (MESa eq 51/52 + SAG 1997); explicit geodesic-branch reduction and epsilon.1 admissibility theorem not in accessible literature
G-MESB-TRACE	C	MESb non-geodesic coefficient trace: registered (3/4,2,2/7)/(3/4,1,3/14) in no accessible source; companion cap max(eps2,eps3)	documented discrepancy against accessible primary sources
G-KE	K+P	King-Ellis rotating-congruence program: contracted Gauss identity, tilt vorticity classification, DeltaOmega.k=0 double obstruction (dynamical)	King & Ellis 1973 base known; the dynamical upgrade of the lower-endpoint W2 withdrawal within the group-invariant perfect-fluid class not found
G-OMK	K+P	Omega.k slaving theorem Sigma = kappa K, kappa = -1/(2+q) exact; near-FLRW kappa(w), c2(w); resonance family w_n	center-manifold tilt dynamics known (Collins-Ellis; Hewitt-Wainwright); exact slaving coefficients with certified interval remainder not found
G-OMK-REM	P	interval-arithmetic certified remainder tube —Sigma - (kappa K + c2 K^2)— j= —K—^3 on omk_domain.v1	branch-and-bound exact-rational forward-invariance certificate; no cosmology equivalent found
G-EGS	K+P	one-way EGS/FLRW theorem + counterexample registry (x.C = 0 never implies FLRW)	EGS theorem known; the sealed converse-refusal counterexample registry is program-specific
G-NT2	K	NT2 reciprocal coefficient bracket and tail convergence separation	elementary algebra/series bounds
G-NTA3	K	sqrt(2/5) estimator-specific dispersion, dof 2l+1	chi-square moment identity
G-ODDL	K	odd-L diagonal BiPoSH structural zero A^LM.ll = 0 for odd L	CRAG A4: known exchange symmetry (Book-Kamionkowski-Souradeep 2012); independently re-proven by sympy Wigner-3j oracle
G-RICCI11	K+C	11-type exact spatial Ricci scalar identity, Koszul engine == Ellis-MacCallum anchor	Ellis-MacCallum 1969 canonical tables known; dual-engine gap 8.2e-10 is implementation cross-check
G-PARITY	K	four exact parity identities; EB/TB ratio parity-even; signed-component template requirement	Pontzen-Challinor 2007 confirmatory; parity-evenness elementary once stated; internal corrective record
G-BPROJ	K	premise-complete axisymmetric B-projector invariant: sigma.20-only + zero B-tower implies exact zero	elementary selection-rule algebra; premise-conditional statement
G-M4B	K	unsigned comparator leakage certificate M_max(B) = 4B, algebraic-only	elementary bound; algebraic-only status part of the statement
G-SWRAY	K	SW-only photon-momentum transport closed form on prescribed Bianchi I kinematics + linear parity/shear quadrupole oracle	conserved-momentum closed form standard; numerical mechanics verification
D-CF4-MANIFEST	C	row-complete authenticated CF4 column manifest, 38053 groups, typed columns	byte-authenticated against the published VizieR J/ApJ/944/94 tables
D-CF4-MV	C	CF4 minimum-variance bulk-flow reproduction: amplitude within a few percent of the published value, apex within 8.5 deg at the inner depth (amplitude headline withheld in favour of the full-covariance treatment)	reproduces Watkins et al. 2023 419 +/- 36
D-CF4-DEFL	C+P	full-covariance significance deflation: 1 sigma honest (cosmic-variance dominated) vs 8 sigma noise-only formal	parallels Whitford et al. 2023 uncertainty-underestimation finding; analytic estimator-matched full-cov treatment as primary not found published
D-CF4-FWD	K+C	Cholesky full-C.ab forward simulator + independent FFT reference; stressor-resolved coverage degradation	GRF mock methodology standard; two-generator normalization cross-check internal
D-CF4-IDSET	P	nuisance-box identified sets per depth shell: bounded at every shell, widening with depth; monopole-protected flow estimand	CRAG A11: no published partial-identification treatment of bulk-flow nuisances
D-CF4-GROWTH	C	depth-resolved fsigma8: nearest shell 0.386 (pull - 0.48 vs comparator), deeper shells unconstrained; joint same-data covariance measured	consistent with CF4 pairwise fsigma8 0.36-0.38 (Courtois et al. 2023)
D-CF4-VCORR	C	field-level ML growth-rate lane agrees with the published CF4 pairwise range; numeric headline pending independent covariance validation	agrees with CF4 literature 0.36-0.38 and Planck-side 0.44
D-CF4-RECON	C	reconstruction-method dependence retained as a systematics design axis; numeric instantiation withheld pending independent validation	design lane against external reconstructions (Carrick, CORAS, Lilow-Veena-Nusser)
D-CF4-MOCKSIG	C	linear GRF + super-sample mock null reproduces the analytic linear significance; the reduction to literature values is the un-modelled nonlinear residual	consistent with Whitford et al. 2023 non-linear underestimation mechanism
D-CF4-CURL	C	WF mean-field curl/div = 0.009 on the real 3-D CF4++ field: potential-flow no-go confirmed	linear-theory potential-flow expectation verified on data
D-K1-CONV	K	canonical K1 convention: frame/alm/mask/proc-nside contract verified on real SMICA/Commander with rotation cross-check	implementation-grade convention verification
D-K1-GLOBALP	C+P	look-elsewhere global p = 39/1000 = 0.039 over six registered statistics under the SMICA-processed FFP10 E2E null (noise-reuse sensitivity 0.033-0.039)	consistent with Planck 2018 VII 2-3 sigma anomaly band; exchangeable pooled-rank estimator composition not found published

D-K1-BOOST	P	zero-parameter boost-BiPoSH residual: observed $L=1$ dipole-frame excitation rank $p = 0.854$ inside the boosted FFP10 null	CRAG A2/A3: literature fits boost amplitude at high ell; zero-parameter low-ell residual consistency test on boost-carrying E2E sims not found
D-K1-EVENL	C	even-L diagonal BiPoSH exchangeable ranks under the same E2E null (v10 new analysis B1)	diagnostic consistency analysis; odd-L quotient uses the known structural zero
D-K1-BIPOSH-SI	C	off-diagonal BiPoSH SI measurement on SMICA/Commander, global $p = 0.68/0.65$	consistent with Planck SI findings
D-DESI-DIPOLE	C	DESI DR1 BGS window-corrected number-count dipole $D = 9.49e-3$, clustering-dominated, LCDM-consistent ($p = 0.90$)	adds a BGS low-z data point; consistent-null direction against clustering mocks
D-DESI-EXACTSEL	P	exact-selection per-mock refit machinery on real random density; official-mock significance deferred	survey-conditional per-mock alpha+beta refit design; result deferred pending complete official mock ensemble (data-availability condition)
D-ACT-ISO	C	ACT DR6 low-L mean-field-debiased kappa isotropy $p = 0.35 + 95\%$ UL on excess $L=2..10$ band power $\downarrow 3.28e-6$ (0.47x null)	consistent with release isotropy expectations; leave-one-sim cross-fit mean field at low L method-level
D-ACT-RAWQE	C	raw-QE inputs not publicly available in DR6 release: RDN0 unformable; release-sim cross-fit pooled rank 0.364	documents a verifiable release property
D-ACT-INBAND	C	strict-in-band kappa modulation: no resolved coupling at current MC resolution (rank 102/401, band $L = 41..762$)	consistency diagnostic on the public release
D-JWST	C	17-anchor JWST distance manifest + probabilistic CF4 crossmatch (10 credible); hierarchical covariance forecast: total uncertainty not identified	non-identification certificate against cited JWST anchor set; positional credibility never identity
D-PR176	K	Tsagas div-v/q cross-falsifier: q response unavailable (non-informative); frozen-covariance self-consistency axis failed honestly (coverage 0.9424)	internal falsifier mechanics; dual-axis bookkeeping
D-PR179	C+P	raw-CF4 reconstruction-independent directional cosmography: H-only exact rank 2/20000 CONDITIONAL on selection systematics; all q results withheld by cubic falsifier (corr 0.98)	CRAG A10: lands on the same systematics interpretation as the 2025 forward-model reanalysis of CF4 H0-anisotropy claims; reconstruction-independent falsifier-gated estimand not found published; promotion held by open selection channel (data-availability condition)
D-COMPARATOR-REGION		combined identified region for g and x.C interval from measured Omega.tilt + MES/OMK ceilings (v10 new analysis B2)	first assembled identified-region rendering of the four-component comparator under measured+ceiling inputs; diagnostic only

8 Registered analyses conditional on data completion

The following analyses are fully specified and will run without design changes when their stated data condition is met; none has been started on partial data.

1. **DESI official-mock causal attribution.** When the complete official validation-mock ensemble (1000 EZmock + 25 AbacusSummit) is on disk and authenticated, the exact-selection dipole machinery runs the pre-registered clustering/kinematic/selection attribution with per-mock refits and the survey-conditional null at full resolution. Condition: transfer completion (290/1000 EZmock at the build probe; days-scale remaining).
2. **Cross-survey negative controls and joint covariance** for the directional-cosmography lane, followed by held-out injection discrimination; the H -only conditional rank of Section 5.2 either survives with its selection channel closed or is retired. Condition: item 1.
3. **Second-pipeline low-multipole replication.** The Commander-processed FFP10 end-to-end ensemble would replicate Section 5.1 under an independent component-separation pipeline. Condition: acquisition of the Commander-processed MC set (bandwidth-bounded).
4. **Independent non-author adjudication** of the peculiar-velocity estimator-level findings whose numerical lanes are withheld above; the withheld instantiations either return with validated uncertainties or stay retired. Condition: completion of the independent validation review.
5. **Native anisotropic-transfer stage.** All family-identification, geometry, and handedness tests (including the signed-component template of Proposition 4.10) remain dormant until the native solver line delivers authenticated transfer functions; no diagnostic in this report substitutes for that stage.

9 Reproducibility

The package is generated by `scripts/build_external_audit_report_v10.py` (byte-stable `--check` mode; the PDF is compiled from the shipped TeX). The data figures are generated by `scripts/v10_report_data_figures.py` from sealed result cards only, and the new even- L analysis by `scripts/k1_evenl_biposh_rank_card.py` against the content-addressed compact map cache. Input artifacts (hashes in `MANIFEST.json`):

```
docs/generated/pr150_e2e_pooled_rank.json
docs/generated/k1_global_maxscan_e2e_full.json
docs/generated/k1_evenl_biposh_rank_card.json
docs/generated/pr180_result_card.json
docs/generated/pr145_bulk_flow_report.json
docs/generated/pr147_identified_set.json
docs/generated/pr148_fsigma8_by_depth.json
docs/generated/desi_dipole_mock_card.json
docs/generated/desi_exact_selection_card.json
docs/generated/act_kappa_card.json
docs/generated/pr177_result_card.json
docs/generated/pr179_result_card.json
docs/generated/pr179_response_identifiability.json
docs/generated/mes_geodesic_refreeze_seal.json
docs/generated/mes_branch_registry_seal.json
docs/generated/k5_omega_k_ceiling_card.json
docs/generated/pr131_coefficients.json
docs/generated/pr175_result_card.json
docs/generated/v10_report_figure_manifest.json
docs/audits/v10_web_crag_20260721/tier_evidence.json
```

External-literature verification for the tier ledger was performed by inline web retrieval on 2026-07-21 (interactive research agents were unavailable at build time; the retrieval record is preserved in the evidence ledger).

References

- [1] G. F. R. Ellis and H. van Elst, “Cosmological models,” NATO ASI C 541, 1 (1999).
- [2] J. Ehlers, P. Geren, and R. K. Sachs, *J. Math. Phys.* 9, 1344 (1968).
- [3] G. F. R. Ellis and M. A. H. MacCallum, *Comm. Math. Phys.* 12, 108 (1969).
- [4] A. R. King and G. F. R. Ellis, *Comm. Math. Phys.* 31, 209 (1973).
- [5] C. B. Collins and G. F. R. Ellis, *Phys. Rep.* 56, 65 (1979).
- [6] C. G. Hewitt and J. Wainwright, *Phys. Rev. D* 46, 4242 (1992).
- [7] R. Maartens, G. F. R. Ellis, and W. R. Stoeger, *Phys. Rev. D* 51, 1525 (1995).
- [8] W. R. Stoeger, M. Araujo, and T. Gebbie, *Astrophys. J.* 476, 435 (1997).
- [9] A. Pontzen and A. Challinor, *MNRAS* 380, 1387 (2007).
- [10] L. G. Book, M. Kamionkowski, and T. Souradeep, *Phys. Rev. D* 85, 023010 (2012).
- [11] Planck Collaboration XXVII, *A&A* 571, A27 (2014).
- [12] Planck Collaboration III, *A&A* 641, A3 (2020).
- [13] Planck Collaboration VII, *A&A* 641, A7 (2020).
- [14] D. J. Schwarz, C. J. Copi, D. Huterer, and G. D. Starkman, *Class. Quantum Grav.* 33, 184001 (2016).
- [15] D. Saadeh et al., *Phys. Rev. Lett.* 117, 131302 (2016).
- [16] R. B. Tully et al., *Astrophys. J.* 944, 94 (2023).
- [17] R. Watkins et al., *MNRAS* 524, 1885 (2023).
- [18] A. M. Whitford, C. Howlett, and T. M. Davis, *MNRAS* 526, 3051 (2023).

- [19] H. M. Courtois et al., *A&A* 670, L15 (2023).
- [20] K. Gorski, *Astrophys. J.* 332, L7 (1988).
- [21] J. Hartlap, P. Simon, and P. Schneider, *A&A* 464, 399 (2007).
- [22] G. W. Imbens and C. F. Manski, *Econometrica* 72, 1845 (2004).
- [23] B. Phipson and G. K. Smyth, *Stat. Appl. Genet. Mol. Biol.* 9, 39 (2010).
- [24] A. Vehtari, A. Gelman, and J. Gabry, *Stat. Comput.* 27, 1413 (2017).
- [25] M. S. Madhavacheril et al., *Astrophys. J.* 962, 113 (2024).
- [26] F. J. Qu et al., *Astrophys. J.* 962, 112 (2024).
- [27] P. Boubel et al., *MNRAS* (2025), CF4 Tully–Fisher zeropoint-dipole analysis.
- [28] “No evidence for local H_0 anisotropy from Tully–Fisher or supernova distances,” *MNRAS* 546, staf2048 (2025), arXiv:2509.14997.